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Digitalisation and Innovation for Sustainable Development

An Analysis of Energy Consumption Patterns in SLSO Healthcare Facilities

Group 6

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Disclosure of use of generative AI

Google's Notebook LLM and ChatGPT were used as a literature reviewer and researcher in order to find sources for our literature research.

Abstract

The healthcare sector's energy consumption is driven by continuous operations, medical equipment and indoor climate requirements. Stockholm County's healthcare provider, Stockholms Läns Sjukvårdsområde (SLSO), operates, among others, around 200 facilities but lacks an integrated, data-driven approach to identify unreasonably high electricity consumption. Without normalized metrics, inefficiencies remain invisible and environmental impacts go unaddressed.

Through data analysis, SLSO wishes to identify vectors of improvement by pinpointing weak points in factors such as correlation with weather temperature or other data sets that could cause higher energy consumption. Another important metric to analyse is whether particular facilities are more inefficient than others despite having the same size and patient throughput.

Gathering the data from energy consumption sensors over a span of 3 years, several approaches were taken to define specific metrics that can help identify energy consumption hotspots.

Through temporal analysis, we identified facilities that showed a strong correlation to air temperature changes, leading to a possible lack of insulation or similar issues. The temporal analysis revealed that 55% of buildings had at least one consumption spike over the three years, with winter months making over half of all anomalies. The study also focused on energy classes and showed how E-class clinics would require a single action plan for all whereas D-class clinics need individual assessments.

Beyond analysis, a theoretical exploration looked at the benefits and limitations of additional sensors, landlord cooperation, staff training (with a demonstrated 11.87% savings in trials), as well as a demand response paradigm shift for SLSO's clinics. Digitalization can be a tool for those strategies (in streamlining and intensifying) but our analysis of ICT through the framework of Hilty and Aebishcer revealed the risks they pose on our global resources.

Contents

1 Introduction

1.1 Background

The healthcare sector is energy intensive and responsible for approximately 4.4% of global net CO_2 emissions[1, p. 4]. In addition, more than half of its total climate footprint is derived from its energy consumption, including electricity, gas, steam, and air conditioning systems [2, p. 1]. As public healthcare organizations strive for climate targets, digitalization offers data-driven solutions to optimize usage and ensure sustainability.

Stockholms Läns Sjukvårdsområde (SLSO) operates many of the healthcare and administrative facilities across the Stockholm region. They have collected electricity usage from ∼200 electricity meters in the period 2023-2025 measured on a monthly basis. While the dataset has provided a good understanding of the total energy consumption, it still misses key information about how, where and when that energy has been used.

With no prior studies being done for SLSO the project has many possible ways to study the availability and possible improvements for their facilities.

1.2 Problem formulation

The limitation of the current system is the lack of integration between energy consumption and the building's details such as area in m^2, type and activities, visitor count and the amount of employees. Some of this information could be found in external databases but they're not connected directly to the electricity meters that SLSO are currently using. As a result the energy consumption is measured in absolute terms which makes it difficult to determine whether a given level of consumption is moderate or unreasonably high for a specific facility.

This lack of data granularity limits the ability to identify facilities consuming more energy than needed. It is challenging to monitor and detect unreasonably high energy usage without standardized measurements e.g kWh/m^2 or other comparisons e.g between buildings that have the same type and activities. Furthermore, the lack of real time sensors limits the ability to understand how and when the energy is consumed, and as a result the avoidable energy consumption and unnecessary gas emissions may continue and affect the environmental sustainability goals and misusing the public's resources.

Meanwhile, many of SLSO's newer facilities already have sensors in place that could be accessed and connected to the energy consumption data. This could provide deeper and early detection of the high consumption root causes. However, this data was not provided and therefore is outside the scope of the current project.

By turning electricity numbers into clear information, the organization can see exactly which buildings are wasting energy and try to improve them.
The central problem of this project is that SLSO lacks insight into whether there is currently any unreasonably high energy consumption in their facilities and how this contributes to unnecessary environmental impacts.

This project aims to bridge this gap by combining available data sources to create a clear, actionable energy analysis model, as detailed in the following sections.

2 Aim and objectives

2.1 Aim

The environmental impact of healthcare is often overlooked as healthcare's importance is very prominent in our society. The aim of this paper is to identify weaknesses in specific sections of healthcare in terms of unreasonably high energy consumption of SLSO's clinics in Stockholm.

This paper also aims to further speculate what potential data could be gathered and what systems could be implemented to give a better overview of energy consumption and further improve the sustainability of the healthcare sector. This will hopefully highlight future research opportunities and suggest strategies that SLSO can pursue to contribute to more sustainable business practices within their facilities.

2.2 Objectives

2.2.1 Theoretical analysis and research

The problem will also include an inquiry into the possibilities for using additional data sources, such as sensor measurements from a building management system, or from information on the degree of occupancy, the climate inside the buildings or how a paradigm shift in energy management could potentially improve energy inefficiency detection even further.

2.2.2 Statistical analyses for high consumption and spikes

This method will focus on identifying buildings that could be consuming unusually high amounts of electricity. The objectives of this method are:

- Detect consumption anomalies: using z-score analysis to identify statistically significant spikes.
- Identify temporal patterns: by analysing spike frequency across months and years, we can detect seasonal variations from real anomalies.
- Compare energy class performance: by evaluating spike frequency for each energy class (A-G).
- Visualize anomaly distributions: by using heatmaps we identify problematic buildings and periods.
- Persistence analysis: detect buildings with consistently high electricity use over multiple months/years.
- Trend detection: highlight buildings whose electricity consumption increases over time.
- Temporal flagging: using normalization to find each building's peak usage months and identify unusual behaviour.

As for the limitations, this method does not consider the buildings' characteristics such as size, area, type (outside of the energy class) or activities. This method focuses on statistical detection of abnormal consumption and cannot directly prove physical inefficiency. The method focuses far more on correlation rather than causation, as causation would require far more data points and is a complex system far outside of the scope of this project.

2.2.3 The impact of weather

To see if power use is driven by heating needs or cooling needs, we can try to know if there is a relationship between weather (via outside air temperature) and energy consumption. Then, the strength of this tendency needs to be quantified. We must also measure how variable the energy consumption is over time, building by building. Both of these metrics can help us pinpoint weather-affected clinics.

Finally, seeing the dispersion of these metrics discriminated by energy-class is a first step towards understanding how efficient a high energy-class (A, B or C) is at preventing energy use from being affected by the weather.

3 Project Design

3.1 On making a temporal analysis of energy use

Combined with other open datasets, we can make estimates of energy efficiencies of SLSO clinics by looking at them over time.

3.1.1 Statistical analyses for high consumption and spikes

This analysis' main goal is to identify irregular electricity consumption patterns in the buildings by combining statistical anomalies detection and data visualization. The process structure that has been used to achieve it is as follows.

1. Data structure: The building electricity consumption data contains 185 buildings, each has $0-36$ data points in (kWh) from January 2023 to December 2025, along with metadata such as meter id, address and energy class (A-G).

2. Statistical normalization: For each building we will calculate:

- Mean consumptions ($\overline{x}$): average monthly electricity usage, where n is amount months with electricity consumption more than 0.

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- Standard deviation (σ): consumption variation using sample standard deviation.

$$\sigma = \sqrt{\frac{\sum (x_i - \overline{x})^2}{n-1}}$$

3. Z-scores: Converting each monthly consumption to a z-score representing its deviation from the building's normal consumption.

$$z_i = \frac{x_i - \overline{x}}{\sigma}$$

With this standardization we can compare the buildings that have a large difference in consumption.

4. Detecting anomalies: in these analyses we define statistical anomalies (spikes) when the $z \geq 2$, so values that are more than two standard deviations above the mean. Statistically speaking this stands for approximately 2.3% of the approximate normal distributed data.

5. Analysis process flow: the python script used to process, analyse and generate the graphs can be found on the project's Github repo and our analysis flow is described in Figure1.

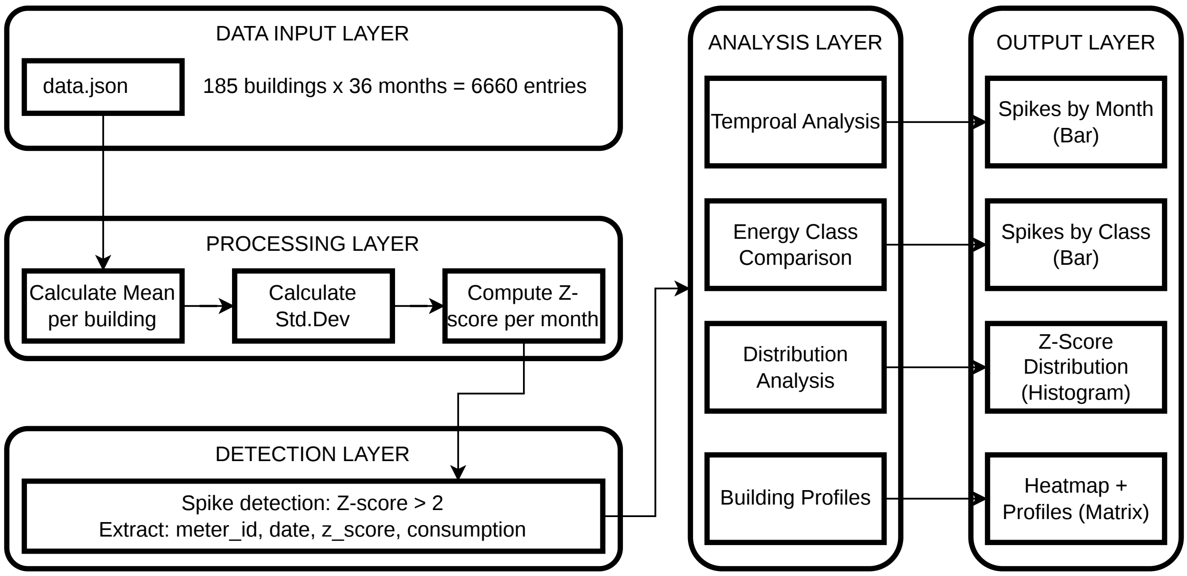

Figure 1: Analysis process flow using Z-scores

3.1.2 Persistence (top 10%)

We compute:

$$I_{i,t} = \begin{cases} 1, & \text{if } z_{i,t} \text{ is in top 10\% of all z-scores} \\ 0, & \text{otherwise.} \end{cases}$$

$$S_i = \sum_{t=1}^{36} I_{i,t}$$

This is essentially a persistence test that determines whether the high consumption occurs frequently rather than randomly.
Buildings with $S_i \geq 6$ months, will be considered as high consumers, indicating consistent high consumption but without being extreme. Buildings with $S_i \geq 8$ are highlighted as statistically unusual with probability of less than 1% under random expectation. This test can eliminate other factors such as area and occupancy, because the class mean and standard deviation capture the typical consumption ranges, so large buildings is only flagged if it really has high consumption not just because it's big. If it was random then

$$E[S_i] = 36 \cdot 0.10 = 3.6$$

but we set the threshold to 6.

Limitations: This normalization only considers differences in the averages and spread of efficiency classes and not the precise physical size of a building. Therefore, buildings that are not typical may still be biased if the class contains too few buildings or high variance.

3.1.3 Temporal scaling and peak identification

Comparing energy use across different buildings can be tricky, since their electricity consumption size is different. A large clinic will naturally use more electricity than a small clinic, which makes it difficult to compare their behaviours using raw numbers. To solve this, a **Min-Max Scaling** technique was used to convert the focus from how much energy is used to when the highest usage happens. This technique makes the comparison fair for all buildings regardless of their size.

Data normalization
To normalize the data, a rescaled formula is used for each month electricity consumption (x) for each building to a range between 0 and 1.
We compute:

$$x'_{i,t} = \frac{x_{i,t} - \min(x_i)}{\max(x_i) - \min(x_i)}$$

Where:

- $x'_{i,t}$:normalized value for building i in month t.
- $x_{i,t}$: is the raw electricity consumption.
- $min(x_i))$: the minimum consumption values a specific building ever achieved over the 36 data points.
- $max(x_i))$: the maximum consumption values a specific building ever achieved over the 36 data points.

The value 1.0 is assigned to the month with the highest electricity usage for each specific building. This value acts as a **flag**, to mark the month each building reached its peak. By doing this we can see the timing of these spikes without being distracted by the actual size of the building or how many kWh it consumes.

Operational application
The main idea of identifying these temporal flags is to isolate buildings with unusual timing, which are buildings that reach their highest energy use in months when no other buildings do. At the same time it helps to identify the months that most buildings reach their highest use of electricity, which shows the general pattern for the electricity usage. Identifying these unusual timing cases helps to find any problem. This method allows to notice any issues that might be missed if the focus was only on the total electricity numbers.

3.1.4 Air temperature to energy consumption correlation

Governmental datasets provided by SMHI observatories can give us monthly readings in several locations in Stockholm of air temperature. We have chosen the Bromma airport observatory dataset since it matches our time frame. [3]

To determine a quantifiable relationship between weather variables and electricity consumption in clinics, our primary analysis is to find a Pearson correlation coefficient between monthly consumption and monthly average temperature for each clinic. This will identify if heating or cooling needs drives power use: negative coefficients indicate consumption decreases as temperature increases (the building's heaviest load is for heating) and vice versa. Moreover, finding buildings with a Pearson correlation coefficient nearing zero suggest successful management and architectural work. To quantify the strength of these tendencies we can compare the coefficients' values.

In practice, this meant aggregating SMHI's hourly temperature readings into monthly averages:

$$\bar{T_m} = \frac{1}{n_m} \sum_{h=1}^{n_m} T_{h,m}$$

Then, for each meter (ie clinic), we calculate the correlation between its monthly consumption C_i and the monthly average temperature $\bar{T}$:

$$r_i = \frac{\sum_{m=1}^{n} (C_{i,m} - \bar{C_i})(\bar{T_m} - \bar{T})}{\sqrt{\sum_{m=1}^{n} (C_{i,m} - \bar{C_i})^2 \cdot \sum_{m=1}^{n} (\bar{T_m} - \bar{T})^2}} \in [-1, 1]$$

Finally, the standard deviation measures the dispersion of monthly consumption values:

$$\sigma_i = \sqrt{\frac{1}{n-1} \sum_{m=1}^{n} (C_{i,m} - \bar{C_i})^2}$$

We have done these computations using Python's pandas.DataFrame library and its functions `corr` and `std` and Matlab for plots and graphs.

Justification of statistical method
The Pearson correlation coefficient measures linear dependence between two continuous variables allowing us to compare facilities regardless of their absolute scale[4, p. 61]. Since energy use in heating-dominated climates (such as Scandinavia) typically shows a linear response to temperature changes, this method finds a metric for weather sensitivity without confusion brought by building volume.

3.1.5 Energy types to weather-dominated correlation

From this correlation work, we attempt to see if energy types play an active role in stabilising electricity consumption at extreme outside temperatures. We can start by breaking down the distribution of correlation coefficients and then standard deviation by building energy types (B, C, D, etc).

3.1.6 Spotting faulty buildings to immediate action

The analyses above work diagnostically: they identify buildings where energy performance deviates significantly from expected patterns and when. Immediate actions could be:

1. Manual control audits: managers should check for equipment that could have been inadvertently left in manual control and thus bypass energy-saving automated schedules.

2. Sensors could be recalibrated: thermostats and controllers require semi-annual recalibration to maintain accuracy. [5, p. 188]

3. Hardware repair: fixing leaks, keeping routine component care (changing filters, cleaning coils and lubricating motors)

4. Alarm analysis: modern systems can give custom alarm messages that both flag a fault and give repair instructions. [5, p. 320]

5. Schedule optimization: setting time schedules so equipment runs only when required. [5, p. 313]

6. Occupant control limits: setting narrower bands on user-adjustable thermostats to prevent users from setting cooling to extreme and energy-intensive levels. [5, p. 314]

For systemic upgrades, SLSO should consider deep renovation work: insulating buildings and replacing old windows. This project will investigate how digitalization can further this action plan (see section 5.2).

3.2 Theoretical part

3.2.1 Study design

The theoretical aspect of the report is a comparative theoretical analysis of how SLSO could optimise their energy usage to reduce their environmental impact and their operational costs. Since this section is mainly theoretical the analysis will be based on previous empirical studies that illustrate the potential energy savings and will apply those findings to the case of SLSO's buildings.

A binary approach to occupancy would not be as fruitful as an occupancy count estimation approach, since SLSO has general high flow of patients throughout the year and a typical day would include over 17 000 patient visits[6]. If occupancy was then reduced to only being occupied or unoccupied one would logically speaking have a high percentage of rooms with constant occupancy. This in turn would not be particularly useful for energy optimisation as the granularity would be low and the data would not have the same potential to lead to energy optimisation. Hence, the focus around occupancy will be related to count estimation, and how SLSO could collect the necessary data for such an estimation and how this estimation could be used to reduce the energy consumption.

3.2.2 Data collection approach

SLSO does not own the buildings they operate their clinics from. They rent around half of them from Locum and the other half from other landlords. The mix of landlords combined with lack of permission on insights into the data collected results in this part being based on what sensors would be ideal to have in each SLSO occupied building.

3.2.3 Occupant counting sensor

Research demonstrates the possibility of using occupancy sensors to control HVAC systems to decrease the energy consumption by anywhere from 20 to 40 % while not compromising on comfort in the offices[7, p.17]. These strategies could be put in place to decrease the energy consumption further and improve air quality without major structural changes, this is a process known as retrofitting[8, p.208].

4 Results

4.1 Temporal analysis

4.1.1 Statistical analyses for high consumption and spikes

In the z-score analysis of our 185 buildings over 36 months (January 2023 to December 2025) we identified 158 consumption spikes where $z \geq 2$. These spikes were caused by 102 of 185 buildings about 55% of the dataset.

1. Z-score distribution: the histogram of all z-scores in Figure 2 confirms that the consumptions follow an approximately normal distribution, where:

- Majority of consumptions fall within $-1 \leq z \leq 1$, representing normal consumption behaviour.
- 158 observations has $z \geq 2$, which correspond to around 2.4% of total data points.
- This aligns with our statistical expectations assuming it's normally distributed data.

As a result the distribution validates our assumption and confirms that the flagged 158 spikes can be unreasonably high consumption events.

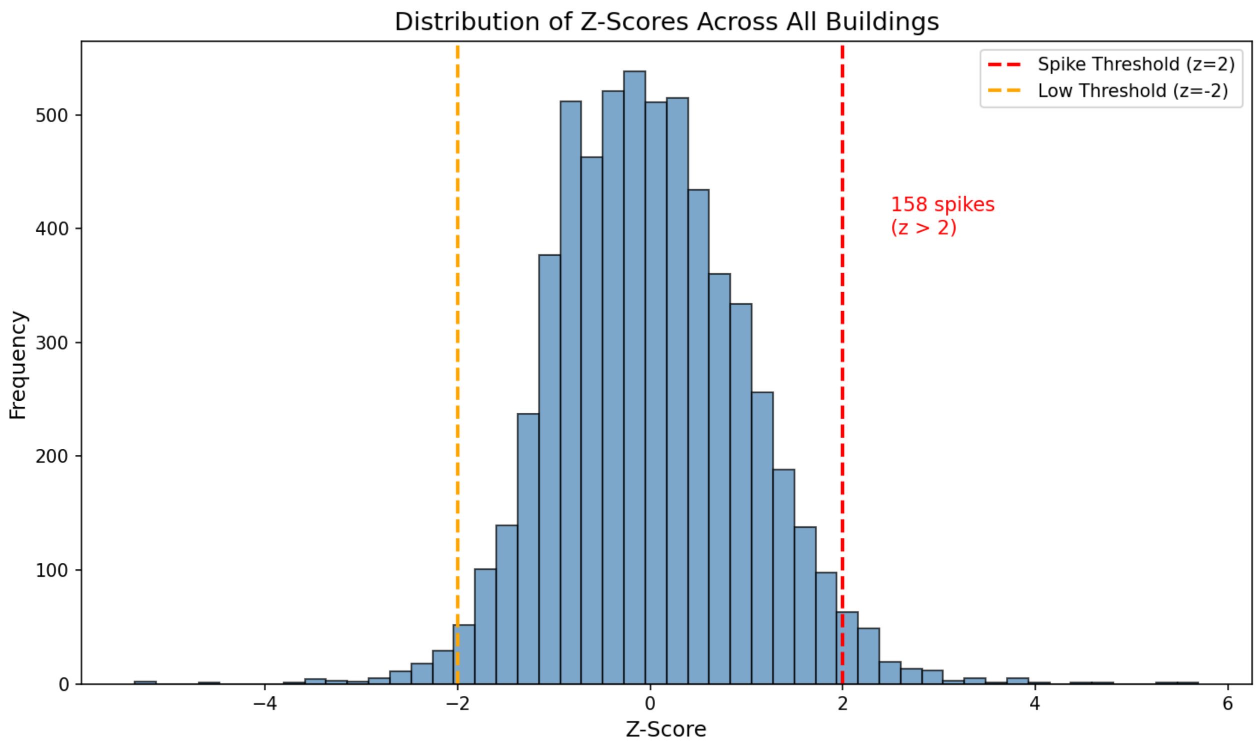

Figure 2: Histogram of all z-scores with threshold lines

2. Temporal pattern analysis: the monthly spike results (see Table 1) show that a strong seasonal pattern in consumption anomalies.

Table 1: Monthly spike distribution by season

Season	**Months**	**Total Spikes**	**Percentage**
Winter	Jan, Feb, Dec	80	50.6%
Spring	Mar, Apr, May	28	17.7%
Summer	Jun, Jul, Aug	14	8.9%
Autumn	Sep, Oct, Nov	36	22.8%

As shown in Figure 3, January clearly dominates with **54 spikes** around 34% of all spikes, it's most likely caused by heating which is needed during cold periods in Sweden. On the other hand, summer months show minimal spikes, where June had only 1 spike during these three years. This can confirm that most of the detected spikes correspond to heating purposes rather than equipment malfunctions or other consumptions like lighting or ventilation.

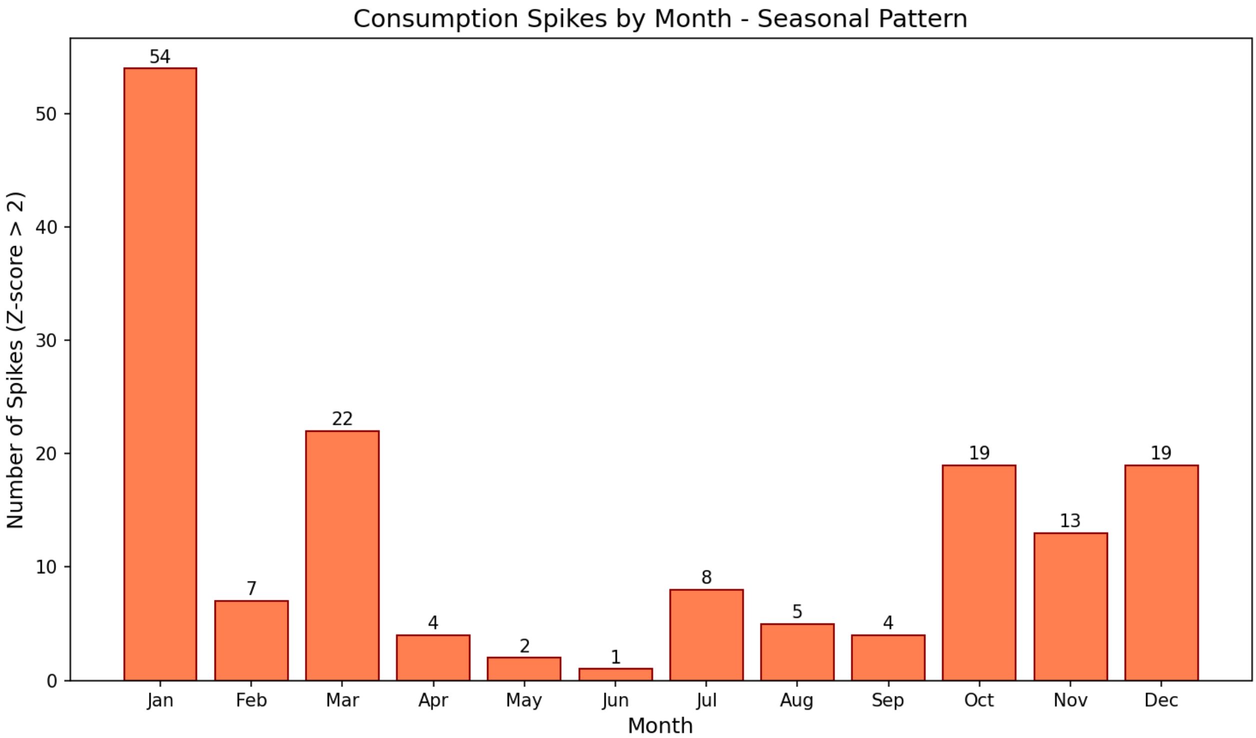

Figure 3: Seasonal pattern - shows which months have most spikes

3. Energy class analysis: Figure 4 shows how buildings that have E Class also have the highest spike rate of 1.05 spikes per building which suggests that buildings with low energy class are most likely to consume more energy and implies higher chance for spikes. On the other hand buildings with B class have the lowest rate of 0.69 with more stable consumptions patterns. Buildings with "unknown" energy class need to be researched after collecting the needed data, but these could be distributed among other classes with the current class ratio.

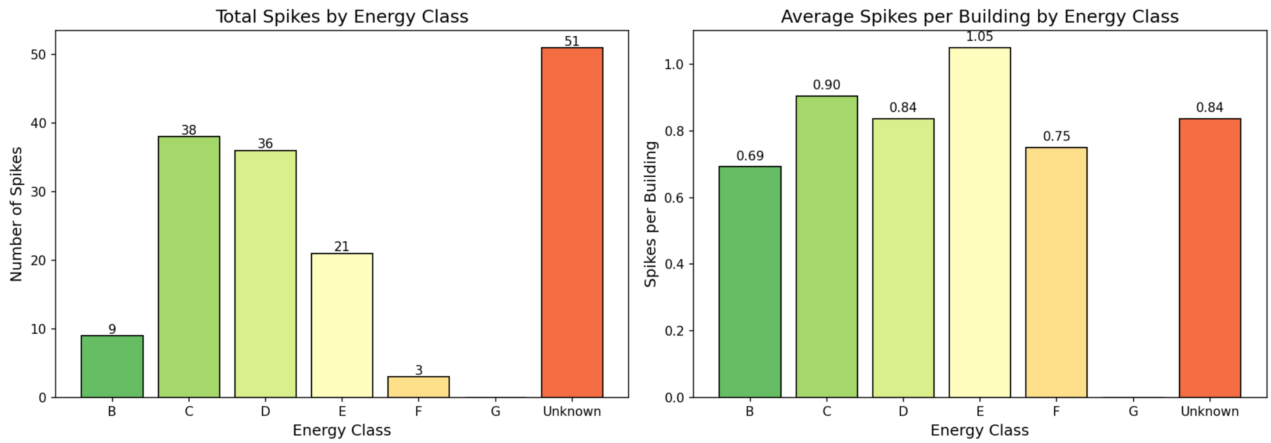

Figure 4: Energy class comparison - absolute counts & per-building rates

4. Spike heatmap analysis: Figure 5 shows a heatmap of all 102 buildings with spikes against their 36 data points. Table 2 links observed clusters in the heatmap with their most probable cause.

Table 2: Clusters and patterns in the heatmap and their cause

Pattern Type	Observation	Cause
Horizontal lines	Dense red clusters in January in all 3 years	Winter conditions.
Isolated intense cells	Random spikes with $z > 4$	Potential equipment malfunctions.
Vertical lines	Multiple spikes on different months of the same building	Persistent consumption instability.
White regions	June-August rows are mostly with no spikes	Stable summer consumption patter.

Extreme events worth investigating fall above the typical seasonal variation and may be caused by equipment malfunctions, unexpected event or other errors. They are listed in Table 3.

Table 3: Extreme consumption anomalies by facility with Z-score above 4

Meter ID	Date	Z-Score	Energy Class
132	September 2023	5.70	C
119	April 2025	4.79	B
173	September 2025	4.06	Unknown
315	January 2023	4.49	Unknown

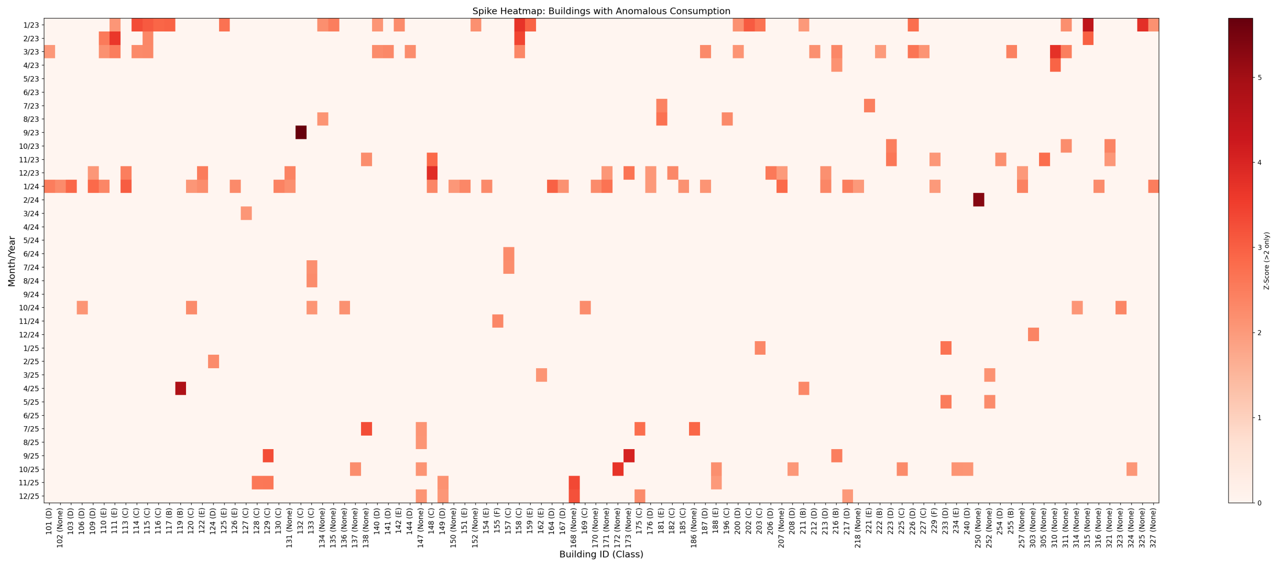

Figure 5: Heatmap showing buildings vs time with spike patterns

4.1.2 Persistence top 10%

The analysis shows 21 buildings with consistently high energy consumption, appearing the top 10% for 6 months or more. Among these, 6 buildings are highlighted as extreme cases being in the 10% for 8 months or more (see Figure 6).

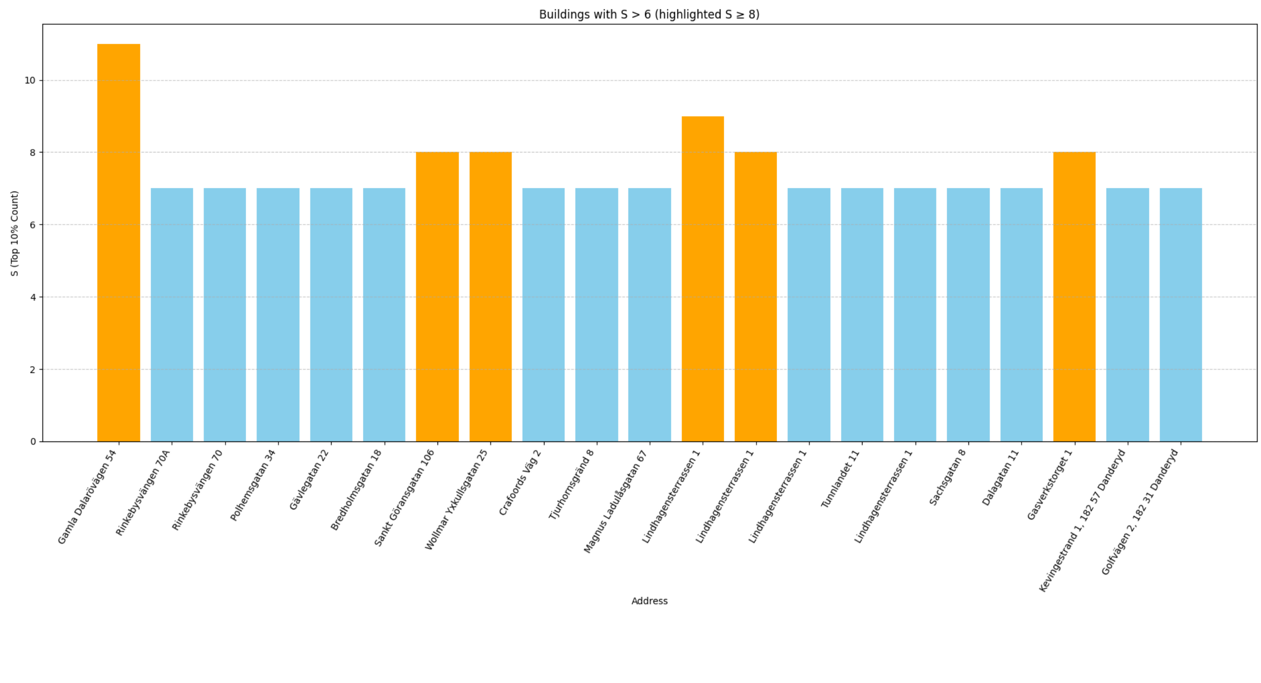

Figure 6: Persistence - shows buildings with consistent high energy consumption

4.1.3 Temporal scaling and peak identification

The temporal analysis began by looking at when all buildings reached their highest electricity use over the three years. Figure 7 shows a clear pattern, The majority of the 200 buildings reach their maximum electricity consumption (Flag 1.0) during the winter months, around December and January. This makes sense because, in Stockholm, buildings need the most energy for heating and lights during these cold and dark months. This gives a normal pattern to compare everything else against.

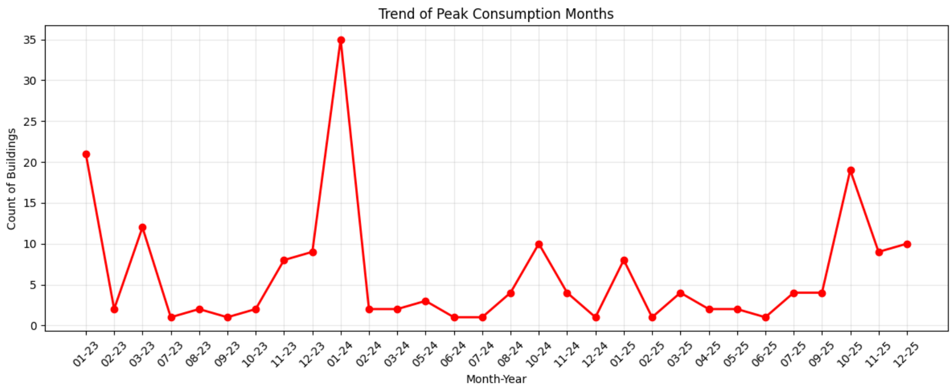

Figure 7: Monthly distribution of peak energy consumption flags across the buildings

After understanding the general trend, a granular analysis was performed for each building as shows in Figure 8. These plots show exactly when each specific building reached its peak consumption. This step is important because it allows us to see the specific timing for every building to help to check if a building follows the expected schedule or if it has an unusual timing for its maximum energy use.

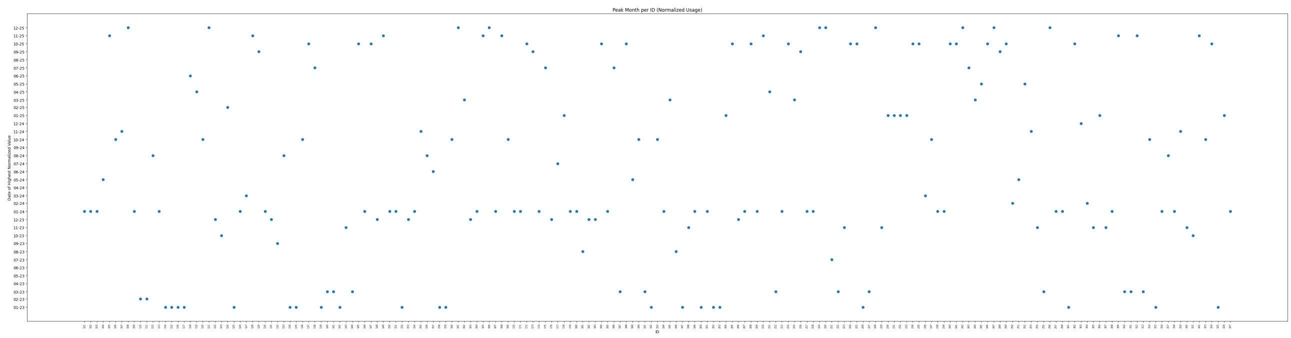

Figure 8: Individual building profiles showing normalized energy peaks over time

The final stage of the temporal analysis is the identification of operational anomalies as visualized in Figure 9. This dashboard acts like an alarm system. While the blue areas show the general data, the red exclamation marks are the critical part where these marks highlight buildings that hit their peak (Flag 1.0) energy use during unusual periods, such us during the summer. These flags indicate a negative correlation with typical seasonal heating demands. By isolating these specific cases, the model figure gives a prioritized list of buildings that need to check first for any issues like poor cooling efficiency or wrong control settings.

4.1.4 Air temperature to energy consumption correlation

Figure 10 shows the distribution of Pearson correlation coefficient and standard deviation in the outside temperature to electricity consumption correlation. Subplot a represents SLSO's

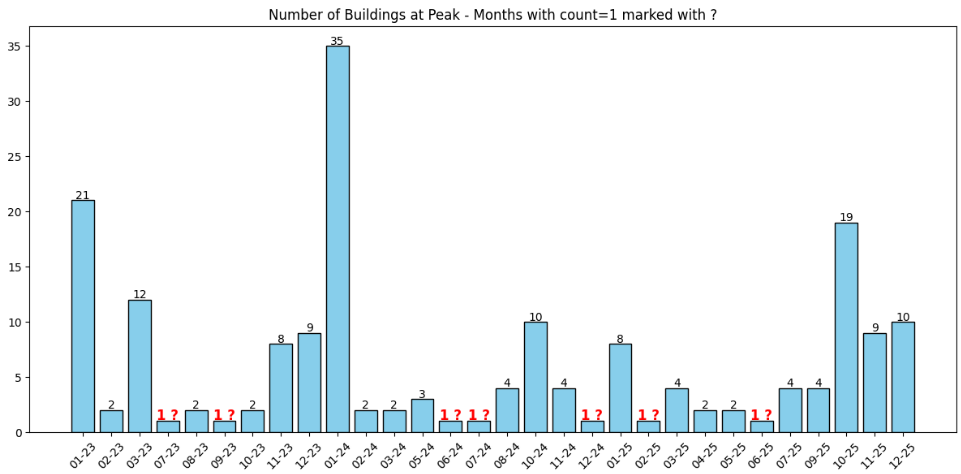

Figure 9: Identification of facilities peaking outside typical winter season

clinics tendencies to have negative correlations with a peak at around −0.7. This means that the majority of buildings consume more electricity when temperatures are low for winter heating. This is to be expected for Stockholm. Thus heating is a major force of electricity consumption for SLSO. Subplot b shows how much electricity consumption varies month to month across buildings. The data shows that the distribution is wide: clinics are in different types of buildings and some have extremely high variability (located in the right part of the plot). Subplot c depicts both aspects (correlation and variability) into one plot. The top-left corner gathers clinics with large and seasonal heating needs which should be SLSO's priority for efficiency audits.

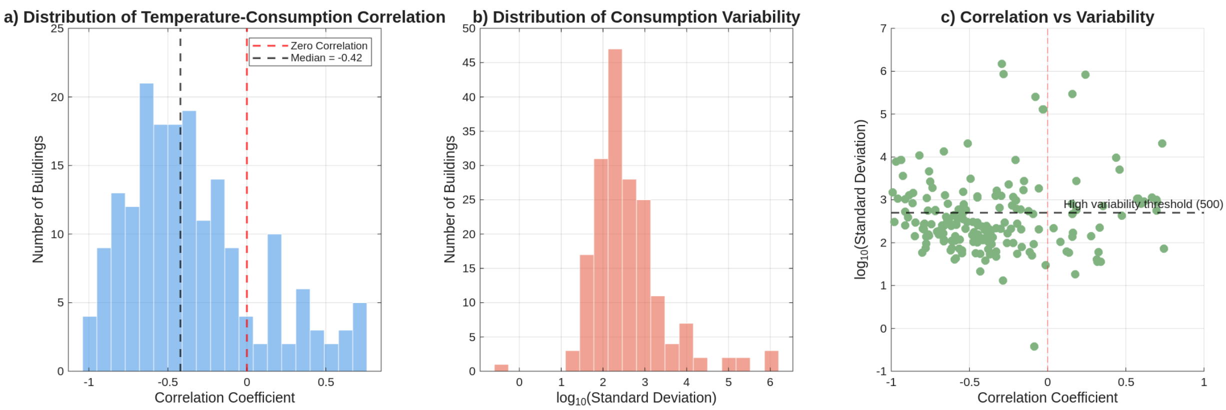

Figure 10: Distribution analysis of outside temperature to electricity consumption correlation

We can also list (see Table 4) the most extreme buildings: those with both strongest correlation coefficient and strongest standard deviation should have priority for energy efficiency intervention.

4.1.5 Energy types to weather-dominated buildings

Figure 11 is the distribution of correlation coefficients and standard deviation broken down by building energy types. Though we have insufficient data to fully exploit it, we can safely say that type E buildings behave similarly to one another because they show strong negative correlation. This means that working on one energy saving solution for one will likely work for all. On the other hand, type D buildings have the most variation: electricity consumption is affected by weather, use patterns, equipment and maybe other variables. Consequently,

Table 4: 10 worst weather-dominated SLSO clinics

Rank	Address	Corr	Std Dev	Score
1	Lindhagensterrassen 1	-0.293	1481942.0	1.598
2	Sjukhusbacken 10	-0.281	864903.9	1.175
3	Hantverkargatan 45K	-0.990	1509.2	1.001
4	Norr Mälarstrand 40	-0.979	300.1	0.994
5	Hantverkargatan 45K	-0.970	7852.4	0.994
6	Stockholmsvägen 43, 182 78 Stocksund	-0.959	1072.0	0.983
7	Hantverkargatan 45	-0.940	8391.5	0.977
8	Nykroppagatan 35	-0.935	8401.6	0.974
9	Crafoords Väg 19	-0.926	3644.7	0.966
10	Sibeliusgången 20A	-0.911	521.6	0.955

this group needs individual assessment solutions. Finally, type C buildings are consistently heating-dominated so improvement work should focus on this aspect.

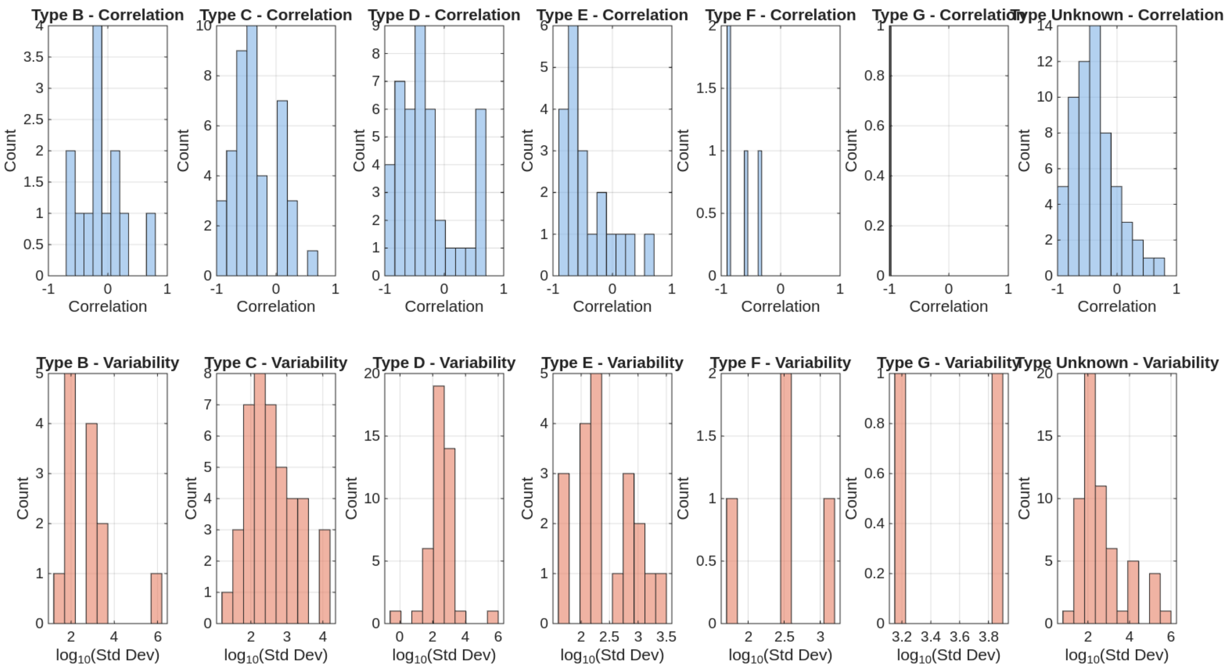

Figure 11: Energy types correlation distribution

5 Discussion and Conclusions

5.1 Understanding the temporal analysis results

The temporal analysis indicates that a significant portion of SLSO's facilities suffer from energy consumption spikes. These spikes have a tendency to correlate to winter seasons and lower energy class buildings. Another point of consideration is the strong tendency of certain buildings to remain in the top energy consumers. This can indicate operational inefficiencies, but this could potentially be explained by a high average visitor count. The persistence analysis found 21 buildings that were consistently in the top 10% of energy users for 6 months or more, and 6 of these were extreme cases. Time-based scaling showed that while most buildings peak in winter, a small group peaks in summer, which suggests issues with cooling rather than heating. Weather analysis showed a strong link to temperature (with a Pearson score around -0.7), confirming that heating is the main cause of high energy use. The best candidates for repairs are buildings that show both a strong link to weather and high variation in energy use. Finally, looking at energy classes suggested that Type E buildings all act the same way and can be fixed with one single plan while Type D buildings need to be checked one by one, and Type C buildings clearly need better insulation and wall upgrades.

5.2 From temporal analysis to active control

Paragraph 3.1.6 has detailed the possible first steps that SLSO could take after diagnosing its clinic pool and identifying weak spots. However, digitalization offers a way of optimizing energy performance in priority buildings (those in the top left corner of subplot c of Figure 10). For them, smart heating controls are a high-impact, low-cost intervention. They need a predictive model: their high variability suggests that current heating systems are reacting inefficiently to temperature changes, while strong correlation confirms weather drives this inefficiency.

Digitalization would, however, require additional data. CO_2 sensors can deliver real-time occupancy readings and sub-metering at room-level which would improve accuracy. It would make categorization of energy use (ventilation, heating, cooling, lighting, plug loads, heavy medical equipment etc) possible. Thus transforming entire buildings into transparent systems.

Two distinct control strategies emerge. The first involves fitting existing heating and cooling systems with smart thermostats that "anticipate the behaviour of occupants (based on past experience and use real-time weather forecasts to better predict heating and cooling needs" [9, p. 42]. The IEA shows such devices "can achieve average energy savings of 10-12% for heating and 15% for cooling. [9, p. 45]. But these devices can struggle with inconsistent routines and conflicting user habits.

Another approach uses predictive models that anticipate rather than react. SLSO could integrate weather forecasts with occupancy patterns (e.g., fewer medical visits midday), then the building's energy management system pre-heats spaces before cold arrives or leverages cheaper electricity during certain periods[10]. This enables real-time adjustment with continuous fine-tuning based on multiple data streams[9, p. 43] and outperforms baseline control to up to 50% energy savings [11, p. 14].

But two cautions need addressing. The first concerns the rebound effect described in section 5.6.1. The second caution concerns human operators. Without adequate training (see section 5.4) and intuitive interfaces, staff might induce worst efficiency than planned due to frustration with an overly complex system.

5.3 Towards demand response participation of clinics

A transformative strategy could come from SLSO managing clinics as an interconnected responsive network rather than isolated consumers. With this mindset, SLSO could modulate its total electricity consumption in response to grid conditions, updating loads to prioritize critical clinics and avoid waste energy usage. Thus each clinic can become an active participant in grid stability. As the IEA explains, such "smart demand reponse" allowes consumers to "alter when they draw electricity from the grid" [9, p. 83]. Thus, SLSO could implement a centralized subnetwork control manager acting as an internal energy aggregator via predictive models to dynamically allocate power and strategically shift loads across clinics, optimizing total consumption while maintaining essential services.

But this introduces significant operational and ethical risks. Firstly, a predictive optimization model might classify certain clinics as non-essential during grid stress events and reduce power allocation (catastrophic if they serve vulnerable, isolated populations). This reflects the principle that sustainable systems must prioritize people and technological optimization must not overwrite human needs[12, p. 512].
Secondly, a centralized control creates an attractive single point of failure. As cybersecurity experts warn, systems with "massively expanding 'attack surface' ", one compromised device can become a "weak point for the whole system" [9, p. 126].

Finally, over-reliance on sophisticated digital control systems may reduce crisis resilience: if the predictive model fails, communications are cut or the central manager is compromised then the entire network becomes unusable. A resilient strategy needs redundant fallback mechanisms, clear ethical protocols for load-shedding prioritization, robust cybersecurity architecture and human oversight.

Consequently, SLSO could implement a consistent secondary grid system for critical electricity needs alongside a predictive controller. Each clinic should identify its critical energy needs (power for all critical equipment and basic HVAC). The secondary system's goal is to ensure that the grid can supply this critical threshold, leveraging the initial "isolated clinics" paradigm to do so.

5.4 Shift in behaviours and its effects

Currently, SLSO does not provide employee energy-saving behaviours (EEB) training but it could benefit from it. Indeed a field trial in the United States showed that employees who received social norms feedback reduced their energy consumption by 10% during the intervention and by 11% at follow-up monitoring. [13, p. 1]

However, educating employees on "how and why" to save energy can prove to be counterproductive as far as a 4% increase in energy consumption [14, p. 1]. That is why SLSO should adopt a mix of leadership, social and technological strategies, as suggested by Yixiang Zhang and Bowen Fu in their metastudy:

1. Tailored training: interventions for office-based administrative staff may differ from those for clinical staff in production-heavy areas[15, p. 11]

2. Role modeling: leaders should act as role models by consistently performing energy-saving actions such as turning off unnecessary medical equipment or office lights

3. Communicating vision: staff training should include a clear "sustainable development vision" from management

4. Peer educators: colleagues who encourage one another is a proven strategy and the most likely to be adopted long term by SLSO staff

5. Social norms: giving staff the data showing how their department's energy use compares to others (social norms feedback) is the strategy with highest energy consumption reduction measured

6. Visual feedback: IoT can be used to create power consumption profiles to help staff understand their own usage patterns

7. Gamified systems: a gamified information system with incentives can significantly enhance staff engagement by turning energy conservation into a rewarding challenge[16, p. 5]

5.5 Possible future research paths for SLSO

5.5.1 Sector comparison

A method to identify inefficient usage of energy could be comparing facilities that share the same sector of healthcare. This could indicate if one healthcare clinic of a certain type is using more or less energy than one of the same type, this could be further analysed if the visitor counts were available for these facilities.

With the data received from the SLSO supervisor this could not be reliably retrieved for each facility, especially considering the possibility of mixed-use buildings and with some of the buildings being administrative or other facilities. What can be considered as a future research methodology to SLSO is primarily disregarding the non clinics in this topic and then doing a comparison based on the type of clinic.

5.5.2 Visitor counts

Another helpful metric would be analysing a correlation between hospital visits to the energy consumption based off the number of visits, This would strongly indicate if any unsustainable practices would lead to an exponential growth in energy consumption if the visitor count were to grow.

5.5.3 Area analysis

Having more descriptive data of each facility such as its size could indicate if certain facilities use a higher energy consumption respective of their size, seeing as smaller buildings should use less energy in comparison to a larger one. This data could not be retrieved from open data due to the strong possibility of mixed use buildings.

5.5.4 Multivariable statistical work: multiple regression analysis and STL

The statistical work presented in section 4.1.4 only looked at two variables: outside air temperature and energy consumption. But with more data (such as visitor count, CO^2 readings, precipitation levels, air quality... etc) a new strategy must be adopted. Multiple regression analysis will isolate the effect of one variable only on the energy consumption readings. And statistical analysis with moving averages along with time series decomposition techniques (such as STL decomposition) will help us consider seasons more appropriately.

In the future SLSO could continue the work from this project and identify correlations to energy consumption or possibly justify the high energy consumption that was identified.

5.6 ICT as a solution and problem

5.6.1 Implementing HVAC control systems

The framework presented by Hilty and Aebischer for analysing ICT as part of the solution and problem is applicable in this part of the project [17, p. 22–23].

Level 1 - Direct negative effects
The direct negative effects reside at the first level of the matrix of ICT effects. In the case of implementing HVAC control systems the direct negative effects will largely depend on the sensors already in place. Information on what type of sensors each building contains is, as previously mentioned, limited and not shared directly with SLSO from their landlords. Therefore, the negative direct effects of a HVAC control system implementation in SLSO buildings is hard to estimate. However, if we assume the worst case scenario where no such sensors are present and make that the basis for this analysis, then one can naturally subtract the negative direct effects of the sensors already present in a building. Using the LCA (Life Cycle Assement) approach to materials, resources and energy consumption, one can find that the sensors and control systems could potentially have significant negative impacts.
1) *The manufacturing phase*: HVAC sensors and control units are more or less purely electronic components. These require precious metals like gold and cobalt, often extracted in social and environmental non-sustainable ways in third world countries, like Ghana, using child artisanal miners [18, p. 6] [19]. Additionally, the electronic components needed also require semiconductors which have an extremely energy and resource intensive manufacturing process [20, p. 1].
2) *The operational phase*: While these sensors are in use they will consume energy. The amount of energy a HVAC system will use is of course dependent on its size, the number of censors and the type of sensors used. However, the energy used will still be noticeable.
3) *The End-of-Life phase*: The electronic components used in such a HVAC system will at the end of their life generate e-waste. E-waste can be recycled, but it has been shown that parts of the e-waste recycling industry have some non-ideal practices where e-waste is shipped of to third world countries where people, by hand, burn and disassemble the e-waste to harvest the precious metals found within [21]. Such a ”recycling” process is very hazardous and not sustainable from an environmental and social perspective.

Level 2 - Enabling effects
The enabling effects can be found at the second level of the matrix of ICT effects. *The obsolescence effect* could be relevant with regard to implementing a HVAC control system as the sensors could have its useful lifetime shortened if the software used to control and communicate with the sensors is not supported for further updates rendering the sensors obsolete. *The optimization effect* is relevant with respects to implementing a HVAC control system as the end goal of such a system is to optimize the use of energy consumed by the building's HVAC system. In an empirical study an occupant-centric HVAC and lighting control system was implemented in commercial office spaces in Shenzhen, China. The 129-day study showed a total of 11.87% reduction in room energy consumption, with also an increase in overall temperature perception [22, p. 1].

Level 3 - Systemic effects
The systematic effects can be found at the third and final level of the matrix of ICT effects. *The rebound effect* could be relevant in terms of implementing HVAC control systems in multiple SLSO buildings. A behavioural rebound effect might be of concern. When the HVAC systems have the potential of reducing the energy consumed the people who previously were trying to be energy efficient and save on power used would then not be as incentivised to continue doing

that. The overall expected energy savings from smart HVAC control systems could potentially be partially cancelled due to this behavioural rebound effect[23, p. 1]. However, in public service buildings, this rebound effect might not be so strong as for residential buildings as a German study suggests. [24, p. 6].
While the primary positive impact of HVAC control systems occurs at the enabling level, through optimization of energy consumption, one could imagine that broader systemic impacts can potentially arise if such systems become widely adopted. These include, amongst other things, improved environmental governance through data transparency and easier integration with renewable energy systems that are already in place.

The path towards environmentally strategic digitalization
The four paths towards environmentally strategic digitalization presented by Höjer et. al., are the paths of informing, replacing, intensifying, and streamlining [25, p. 10]. An OC (occupant-centric) HVAC control system implementation would be on the path of streamlining. This is due to that this type of solution would ”streamline things using technology, which means we waste less resources without actually changing the phenomenon” [25, p. 10].

5.6.2 Data analysis

For the data analysis done in this project the effects of ICT matrix presented by Hilty and Aebischer is not as relevant as the data analysis is not an ICT solution, but rather a pure analysis. One could argue that the path framework presented by Höjer et. al. could be more relevant as this sort of data analysis would help with making better use of already available resources and could therefore be argued is on the path of intensifying. [25, p. 10].

5.7 Conclusion

This report has demonstrated that even with monthly power consumption data and limited building metadata, inefficiencies can be identified through statistical anomaly detection and weather correlation analysis. Our strategy-design approach aligns with Höjer et al.'s path of intensifying, as it makes better use of already available resources[25, p. 10]. The path forward for SLSO involves:

1. Immediate investigation of the high-consuming and oddly-timed buildings identified in the analysis.
2. Strategic implementation of smart controls in the most weather-sensitive and volatile clinics: this continues the path of streamlining (where technology reduces resource waste)[25, p. 10].
3. Long-term engagement with landlords to deploy sensors and building management system integration.
4. Tailored staff training.

Digitalization offers powerful tools for energy optimization, but it must be deployed with careful attention to:

- Their full lifecycle impacts (from resource-intensive semiconductor manufacturing, conflict mineral extraction, to e-waste disposal).
- The human factors behind their usage and effectiveness (with the risk of a rebound effect if behavioural changes accompany technological investment).
- The ethical imperatives behind any system affecting patient care being at stake of major shifts after innovation.

With this approach, SLSO can reduce its environmental footprint, lower its operation costs and help making the healthcare sector more sustainable without cutting down on the quality of care it provides.

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